



SONARGAON UNIVERSITY (SU)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CAPSTONE DEFENSE SCRIPT & VIVA VOCE PLAYBOOK

Project: **ADVANCE TRAVEL: A Modern Web-Based Tourism Management & Automated Booking Platform**

Academic Session: September 2026 • 16-Slide Architected PowerPoint Defense Deck

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1. Presenter Roster & Engineering Task Breakdown

The presentation divides the 16 slides equitably among the three researchers following the official engineering task allocation specified in **Table 3.6** of the thesis book. Total formal presentation time is structured for **15 to 16 minutes**, reserving 10 minutes for examiner questions.

Presenter & Student ID	Assigned Engineering Role	Key Developed Modules (Thesis Table 3.6)	Slides & Time
Md. Amir Shorif Misty ID: CSE2203027142 (Lead Presenter)	Lead Backend Architect & Security Engineer	Laravel 12 enterprise setup, MVC system architecture, Authentication & RBAC middleware, Cryptographic security protocols, Admin refund endpoints	Slides 1–4 (~3.5 Mins)
Md. Abdur Rakib ID: CSE2203027141	Database Architect & Data Systems Engineer	Database 3NF relational schema design, ERD modeling, Concurrency control & row-level locking, Administrative BI analytics, Black-box testing matrices	Slides 5, 10–15 (~5.0 Mins)
Efaz Ahammad Thalha ID: CSE2203027172	Frontend Lead & UI/UX Developer	Responsive Blade layouts, Tailwind CSS design system, Interactive destination showcase & catalog filters, Dynamic booking & passenger roster, Live demo	Slides 6–9 (~4.0 Mins)
All Three Presenters (Joint Delivery)	Capstone Research Team	Thesis conclusions, ecological impact, future works, and open floor defense panel cross-examination	Slide 16 (~10.0 Mins)

2. Slide-by-Slide Speaking Script & Timing Guide

Part I: System Architecture & Problem Context

Primary Presenter: Speaker 1: Md. Amir Shorif Misty (Lead Backend Architect)

Slide 1: Title & Presentation Team Credentials [Estimated: 40s]

Verbal Speaking Script: "Honorable Session Chair, respected internal and external defense panel members, and our honorable supervisor, Md. Rashedul Islam sir, Assalamu Alaikum. I am Md. Amir Shorif Misty (ID: CSE2203027142), Lead Backend Architect. Along with my fellow researchers Efaz Ahammad Thalha (Frontend Lead) and Md. Abdur Rakib (Database Architect), we are honored to defend our B.Sc. in CSE Capstone Project titled: 'ADVANCE TRAVEL: A Modern Web-Based Tourism Management and Automated Booking Platform'."

Slide 2: Background & Problem Statement (Chapter 1) [Estimated: 50s]

Verbal Speaking Script: "Bangladesh's domestic travel planning historically suffers from severe operational fragmentation: travelers must visit separate bus terminals, train counters, and hoteliers. Manual cash transactions cause overwhelming congestion during festive Eid rushes, paper tickets are vulnerable to scalping and counterfeiting, and operators rely on manual paper notebooks with zero auditability. Advance Travel resolves these challenges through an integrated multi-modal marketplace, atomic concurrency locks, automated SSLCommerz cashless payments, and cryptographically tokenized QR boarding passes."

Slide 3: Literature Review & Competitive Benchmarking (Chapter 2, Table 2.1) [Estimated: 50s]

Verbal Speaking Script: "In Chapter 2, we surveyed commercial market solutions including Shohoz, GoZayaan, and traditional agencies. As detailed in Table 2.1, existing domestic platforms focus strictly on transit tickets or high-budget flights without holiday bundles. Advance Travel bridges this gap with a full tour and room customizer, camera-accessible web QR validation requiring zero proprietary scanners, automated user cancellations, and a built-in real-time administrative analytics suite."

Slide 4: Three-Tier MVC System Architecture (Chapter 3, Figure 3.1) [Estimated: 50s]

Verbal Speaking Script: "Figure 3.1 illustrates our classical Three-Tier MVC architecture: The Presentation Tier renders responsive Blade and Tailwind CSS layouts; the Application Logic Tier powered by Laravel 12 on PHP 8.4 executes RBAC middleware ('is_admin'), booking controllers, and SSLCommerz services; and the Data Storage Tier persists a normalized SQLite/MySQL database with full ACID transaction guarantees and live flight API tracking feeds. I now hand over to our Database Architect, Md. Abdur Rakib, to present our relational schema design and normalization."

Part II: Relational Data Engineering & Schema 3NF

Primary Presenter: Speaker 2: Md. Abdur Rakib (Database Architect & Data Engineer)

Slide 5: Database Schema Design & 3NF Normalization (Chapter 3, Figure 3.6) [Estimated: 50s]

Verbal Speaking Script: "Thank you, Misty. Respected board members, I am Md. Abdur Rakib (ID: CSE2203027141). Our persistence layer consists of 14 relational tables normalized strictly to Third Normal Form (3NF). Single surrogate primary keys ('id : bigint') satisfy 1NF and 2NF by eliminating partial functional dependencies. In 3NF, transitive redundancies are eliminated: user profiles and package pricing are decoupled from reservations, with cascading foreign keys safeguarding referential integrity. To demonstrate how travelers interact with this data layer, I now hand over to our Frontend Lead, Efaz Ahammad Talha."

Part III: Frontend Architecture, Booking Engine & Live Demo

Primary Presenter: Speaker 3: Efaz Ahammad Talha (Frontend Lead & UI/UX Developer)

Slide 6: Customer Experience: Destination Exploration & Catalog (Chapter 4, Figs 4.2 & 4.4) [Estimated: 50s]

Verbal Speaking Script: "Thank you, Rakib. Respected board members, I am Efaz Ahammad Talha (ID: CSE2203027172). On the client side, our priority was eliminating cognitive friction. As shown in Figures 4.2 and 4.4, travelers can explore curated domestic tours with real-time multi-modal filtering across luxury buses, express trains, flights, and holiday tours. Destinations like Cox's Bazar, Sylhet, and Sajek Valley are showcased through responsive grids engineered with Tailwind CSS and Boxicons."

Slide 7: Interactive Tour Booking & Reservation Interface (Chapter 4, Figure 4.5) [Estimated: 50s]

Verbal Speaking Script: "When a traveler initiates booking (Figure 4.5), the dynamic reservation engine takes over. The interface dynamically compiles passenger demographic rosters on the client side without page reloads. The traveler selects dates, room types, and passenger seats, while an automated promo code engine applies instant discounts (such as 'EID2026' 15% discount) with transparent fee calculations."

Slide 8: Traveler Self-Service Dashboard & Trip Actions (Chapter 4, Figure 4.8) [Estimated: 50s]

Verbal Speaking Script: "Figure 4.8 illustrates our traveler self-service portal: users can monitor active departure countdown timers, view journey progression statuses ('Pending' to 'Confirmed' to 'Completed'), re-download printable PDF invoices, and execute self-service trip cancellations prior to departure, triggering automated refund requests."

Slide 9: [LIVE SYSTEM DEMONSTRATION] 3-Minute Walkthrough [Estimated: 90s–120s]

Verbal Speaking Script: "We now initiate our live system demonstration running on our local environment ('http://advance-travel.test'). Notice our four sequential milestones: (1) Filtering tours by destination, (2) Reserving seats with live fare computation, (3) Executing an SSLCommerz sandbox payment with asynchronous IPN webhook callback, and (4) Synthesizing the QR boarding pass and validating it on the public gate verification portal. Next, our Database & Systems Architect, Md. Abdur Rakib, will detail our algorithmic implementations, financial settlement pipeline, testing matrices, and engineering standards."

Part IV: Technical Algorithms, Financial Ledger, Admin BI & Standards

Primary Presenter: Speaker 2: Md. Abdur Rakib (Database Architect & Systems Specialist)

Slide 10: Core Algorithmic Implementations & Technical Code Logic (Chapter 4, Section 4.3) [Estimated: 60s]

Verbal Speaking Script: "Thank you, Efaz. Respected board members, in Chapter 4 we formulated three core deterministic algorithms: First, the Dynamic Pricing Algorithm calculating total fare based on seat counts, room surcharges, and bounded promotional caps. Second, the SSLCommerz Transaction Hash Algorithm computing SHA-512 checksums to prevent parameter tampering. Third, the Tokenized QR Verification Logic cross-referencing booking codes and active payment statuses in real time."

Slide 11: Automated Multi-Channel Financial Settlement Pipeline (Chapters 2 & 4) [Estimated: 50s]

Verbal Speaking Script: "We integrated Bangladesh's premier PCI-DSS certified gateway, SSLCommerz, supporting Cards, bKash, Nagad, and Rocket. The platform implements an asynchronous server-to-server Instant Payment Notification (IPN) webhook. Even if a traveler experiences network failure or closes their browser immediately after payment, our server-side IPN listener verifies the cryptographic hash, reconciles the ledger, and transitions the booking status to 'paid'."

Slide 12: Anti-Counterfeit Digital Boarding Pass & QR Verification Portal (Chapter 4, Figs 4.6 & 4.7) [Estimated: 50s]

Verbal Speaking Script: "Upon payment settlement, the system synthesizes an encrypted digital boarding pass (Figure 4.6). It embeds a high-density Quick Response (QR) matrix containing the unique 'booking_code'. Station attendants scan this QR code with any standard smartphone camera to access the public verification portal (Figure 4.7), which instantly validates authenticity and prevents duplicate entry attempts."

Slide 13: Administrative Control Dashboard & Business Intelligence (Chapter 4, Figs 4.10–4.12) [Estimated: 50s]

Verbal Speaking Script: "For agency management, our administrative ERP portal (Figure 4.10) provides executive business intelligence: live monitoring of Gross Merchandise Volume (GMV: 120,270 BDT), confirmed passenger counts, package inventory CRUD controls (Figure 4.11), and booking lifecycle management with direct refund processing (Figure 4.12)."

Slide 14: Quality Assurance, Testing Matrices & Performance Evaluation (Chapter 4, Table 4.1) [Estimated: 50s]

Verbal Speaking Script: "Quality assurance was validated through our black-box test execution matrix (Table 4.1): all 10 formal test cases (TC-01 to TC-10)—spanning authentication, IPN callbacks, tampered payments, and unauthorized access—achieved a 100% Pass rate. Furthermore, Google Chrome Lighthouse audits yielded exemplary performance: 96/100 Performance, 98/100 Accessibility, 100/100 Best Practices, and 95/100 SEO."

Slide 15: Engineering Standards, Washington Accord & Feasibility (Chapter 5, Tables 5.1–5.4) [Estimated: 50s]

Verbal Speaking Script: "Advance Travel strictly complies with IEEE 12207, ISO/IEC 25010, and OWASP Top 10 security standards. As detailed in Tables 5.3 and 5.4, the project maps directly to Washington Accord (BAETE) complex engineering problem-solving attributes (WP1–WP7) and activities (EP1–EP5). Financially, with an academic R&D; budget of 10,000 BDT and an annual commercial cloud budget of 170,900 BDT (Table 5.2), our 10% platform commission model achieves full capital break-even within 8.2 months."

Part V: Defense Conclusion & Viva Voce

Primary Presenter: All Three Presenters: Misty • Thalha • Rakib

Slide 16: Project Summary, Delivered Contributions & Defense Q&A; (Chapter 6) [Estimated: 60s]

Verbal Speaking Script: "Misty: 'In conclusion, Advance Travel delivers a production-ready, full-stack Laravel 12 tourism ecosystem with zero double-booking concurrency and automated financial settlement.' Efaz: 'The responsive frontend eliminates cognitive friction, saving travelers significant booking overhead and replacing physical paper tickets.' Rakib: 'Our normalized database guarantees ACID transactional integrity, while conserving ~150,000 sheets of paper annually. We express our deepest gratitude to our supervisor Md. Rashedul Islam sir, Department Head Prof. Bulbul Ahamed sir, and the respected Defense Board. We are now open for your questions.'"

3. High-Frequency Viva Voce Questions & Model Answers

The following technical questions are highly likely to be posed by internal and external examiners at Sonargaon University. Team members should deliver these structured, confident answers:

Q1: Why did you choose Laravel 12 and PHP 8.4 instead of a MERN (React/Node) or Django stack?

Model Answer: [Recommended Lead: Md. Amir Shorif Misty — Backend Architect]

As analyzed in Section 2.4 of our thesis, Laravel 12 provides an enterprise-grade, opinionated Model-View-Controller framework with out-of-the-box security guarantees: Eloquent ORM (which enforces parameterized PDO queries preventing SQL injection), CSRF token verification middleware on mutating requests, Bcrypt credential hashing, and native session state management. For transactional, financial-grade booking systems where database atomicity and rapid production reliability are paramount, Laravel's architectural maturity and developer productivity significantly outperform the fragmented configuration overhead of unopinionated Node.js setups.

Q2: How does the system prevent double-booking if two users select the same seat simultaneously?

Model Answer: [Recommended Lead: Md. Abdur Rakib — Database Architect]

We manage concurrency through atomic database transactions utilizing pessimistic row-level locking ('SELECT ... FOR UPDATE' via DB::transaction() in Laravel). When a traveler enters checkout, a temporary seat-reservation lease lock is acquired with a 15-minute expiration timestamp. If a second user attempts to reserve the identical seat before payment completion, the query detects the active hold and rejects the concurrent request with an informative notification.

Q3: What happens if a user's browser crashes or network drops during payment on SSLCommerz?

Model Answer: [Recommended Lead: Md. Amir Shorif Misty / Md. Abdur Rakib]

We do not rely solely on the browser redirect (Success URL). Instead, we implemented SSLCommerz's server-to-server Instant Payment Notification (IPN) webhook. Even if the customer's browser closes or network disconnects, SSLCommerz servers dispatch an asynchronous POST request directly to our '/payment/ipn' endpoint. Our controller verifies the 'tran_id', 'val_id', and SHA-512 checksum, cross-verifies that the paid amount matches the database booking total, and automatically transitions the booking status to 'paid'.

Q4: Can an unauthorized person forge the QR code boarding pass to gain entry?

Model Answer: [Recommended Lead: Efaz Ahammad Thalha — Frontend Lead & UI/UX]

No. The QR code does not store plain text booking data; it encodes a cryptographically signed verification URL embedding the unique alphanumeric 'booking_code'. When scanned by a station attendant's camera, our public verification portal queries the relational database in real time, validates that the ticket status is 'confirmed' and 'paid', and flags the ticket as 'Boarded' to block duplicate reuse.

Q5: Prove that your database schema adheres to Third Normal Form (3NF).

Model Answer: [Recommended Lead: Md. Abdur Rakib — Database Architect]

Our 14 relational tables strictly satisfy 3NF: (1) First Normal Form (1NF): All attributes are atomic and indivisible with unique primary keys. (2) Second Normal Form (2NF): All primary keys consist of single surrogate keys ('id : bigint'), inherently eliminating partial functional dependencies. (3) Third Normal Form (3NF): There are no transitive dependencies. For example, in the 'bookings' table, we store 'user_id' and 'package_id' as foreign keys, but do not duplicate customer addresses or tour pricing inside the bookings table.

Q6: How does this project demonstrate complex engineering problem solving under the Washington Accord?

Model Answer: [Recommended Lead: All Three Presenters]

Our capstone project maps directly to BAETE Complex Engineering Problem profiles WP1 through WP7: WP1 (in-depth computing principles in 3NF database normalization and MVC design), WP2 (conflicting constraints of sub-second query latency vs. pessimistic transactional locking), and WP6/EP3 (societal, legal, and environmental impacts under the Bangladesh ICT Act and paperless green computing conserving ~150,000 paper sheets annually).

4. Live Software Demonstration Playbook (Viva Checklist)

Maintain this exact execution sequence for Slide 9's live software demonstration:

Step & Objective	Action & Screen	Key Features to Point Out to Examiners
Step 1: Tour Discovery	Open Homepage (/) and navigate to /explore	- Multi-modal transit badges (Bus, Train, Flight, Tour) - Dynamic division filtering (Sylhet, Chattogram) - Open 'Sajek Valley 3-Day Tour' package details
Step 2: Dynamic Booking	Click 'Book Now' on package page (/booking)	- Dynamic passenger demographic fields (name, age, emergency contact) - Seat & room selector with live fare recalculation - Apply promo voucher EID2026 to show 15% discount
Step 3: SSLCommerz Payment	Proceed to Checkout & launch SSLCommerz Sandbox	- Multi-channel payment options (Cards, bKash, Nagad) - Simulate successful bKash OTP/PIN transaction - Show automated IPN webhook callback updating status to 'Paid'
Step 4: QR E-Ticket	Redirect to Booking Confirmation (/ticket/{id})	- Digital boarding pass with high-density QR code - Click 'Print/Save as PDF' button - Passenger credentials, seat assignments, and status = 'Confirmed'
Step 5: Gate Verification	Open /verify-ticket/{code} or scan with camera	- Instant green 'AUTHENTIC & VALID BOARDING PASS' badge - Sub-second validation latency (<1s) - Prevents duplicate entry attempts
Step 6: Admin ERP Suite	Log in as Administrator (/admin/dashboard)	- Real-time financial analytics (Total GMV: 120,270 BDT) - Package CRUD management (/admin/packages) - Booking lifecycle overrides and refund processing

Local Server Instance: http://127.0.0.1:8000 or
http://advance-travel.test (PHP 8.4 via Herd)

Traveler Demo Account: user@travel.com | Password:
password123

Database Engine: SQLite / MySQL in 3NF Normalization

Admin Demo Account: admin@travel.com | Password:
password123